Constraints on Higgs-charm couplings at CMS

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Charm tagging: telling c jets from b and light jets

Charm signature

c-hadron mass and lifetime lie **between light and b**:
- secondary vertex (SV), like b jets
- but closer to the primary vertex (PV): easily confused with light jets

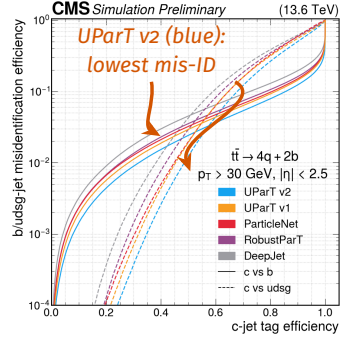


Algorithm evolution at CMS

Likelihood / BDT
JP: track impact parameters
CSV: SV + track

Deep NN
DeepCSV: multi-class b/c/light
DeepJet: constituent

Particle cloud
ParticleNet: graph network



$t\bar{t}$ simulation, 13.6 TeV; solid: c vs b, dashed: c vs light

Direct H → c$\bar{c}$: VH(H → c$\bar{c}$)

Cleanest signature: W/Z → 0, 1, 2 leptons
V+jets and t$\bar{t}$ normalised in control regions

0L: Z→ $\nu\nu$ | **1L: W→ $\ell\nu$** | **2L: Z→ $\ell\ell$**

below — p_T^H = 300 GeV — *above*

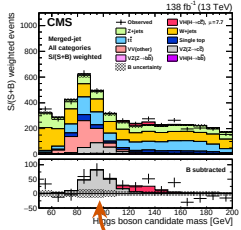
Resolved: two jets

>95% of signal
two DeepJet c-tagged jets
fit to a BDT discriminant

Merged: one large-R jet

<5% of signal, but **less background**
ParticleNet c$\bar{c}$ tag (3 purities)

fit to m(



VZ(Z→ c$\bar{c}$) at 5.7σ validates the method

